If the extracted heat is and external work is , then

The sign “ ” is the ideal limit. In winter, use the refrigerator as a heat pump. At outdoor temperature , it keeps a room at . The warmer room loses heat outdoors. Consider only conduction, governed by the following law. For a conducting layer of thickness , area , temperature difference , the heat conducted through it per unit time from the hot side to the cold side is

Here is the thermal conductivity, determined by the layer material.

. Heat loss is through a pane facing out: in area, thick. The conductivity is ; electricity costs yuan/kWh. Ideally, find the electricity cost of operating the heat pump for .

. Replace it by double glazing: both panes thick, separated by an air gap thick; conductivity . The tariff remains yuan/kWh. For operation, how much is saved versus single glazing?

4.

Fig. has metal , thin insulator and metal , in close contact. Classically, if is a perfect insulator, electrons cannot pass to another metal layer. Yet quantum physics allows this under certain conditions. Tunnelling follows from electron wave nature. Each event transfers one electron. The discrete charge is a multiple of , hence the name single-electron tunnelling. is a tunnelling junction or single-electron tunnelling junction. Coulomb blockade controls it, allowing tiny single-electron devices. This active research field has promising applications.

Fig.

. Treat Fig. as a capacitor of capacitance , with continuous plate charge, as Fig. shows: the electron cloud shifts slightly against the positive background. Tunnelling through the insulator transfers only

Fig.